Power balance and Lawson

Christopher Albert

4.1 Momentum and energy

momentum $\boldsymbol{p}_\alpha = -\boldsymbol{p}_n$

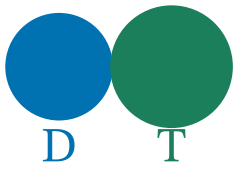
$E = p^2/2m$ $\dfrac{E_\alpha}{E_n} = \dfrac{m_n}{m_\alpha} \approx \dfrac{1}{4}$

$$E_\alpha = 3.5\ \text{MeV}, \quad E_n = 14.1\ \text{MeV}$$

α heats the plasma, n the blanket

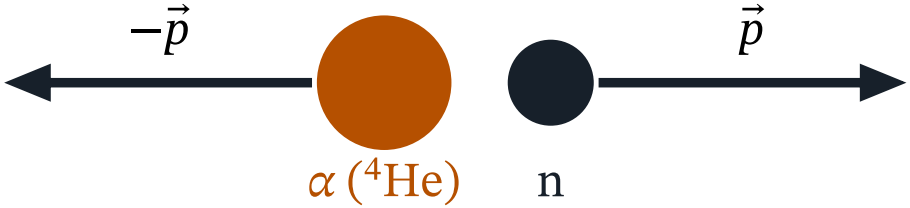
4.1 Alpha and neutron

before



$p_{\text{tot}} \approx 0$

after (centre-of-mass frame)



$$E = \frac{p^2}{2m} \Rightarrow \frac{E_\alpha}{E_n} = \frac{m_n}{m_\alpha} \approx \frac{1}{4}$$

alpha (20 %)

$Q_{\text{DT}} = 17.59$ MeV



with relativistic kinematics: 3.56 MeV and 14.03 MeV

4.2 Fusion power density

fuel $n_\mathrm{D} + n_\mathrm{T} = n, \quad n_\mathrm{D} n_\mathrm{T} = n^2 x(1-x)$

optimum $x = 1/2$

$$p_\alpha = \frac{n^2}{4}\langle\sigma v\rangle E_\alpha, \quad p_\mathrm{fus} = \frac{n^2}{4}\langle\sigma v\rangle E_\mathrm{fus}$$

$n = 10^{20}\mathrm{m}^{-3}$, 14 keV: $p_\mathrm{fus} = 1.7\ \mathrm{MW/m^3}$

4.3 Stored energy and confinement time

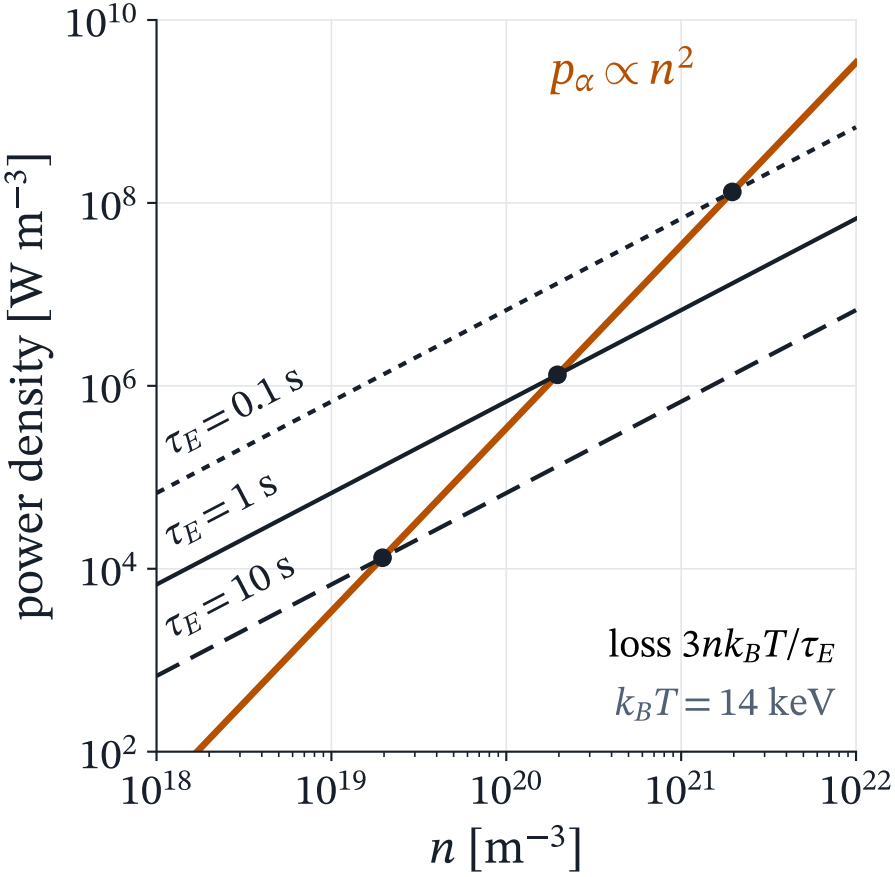
quasineutral $n_e = n_\mathrm{D} + n_\mathrm{T} = n$

stored $W = 3nk_BT$

losses $p_\mathrm{loss} = W/\tau_E$

$$p_\mathrm{ext} + p_\alpha = W/\tau_E$$

steady state

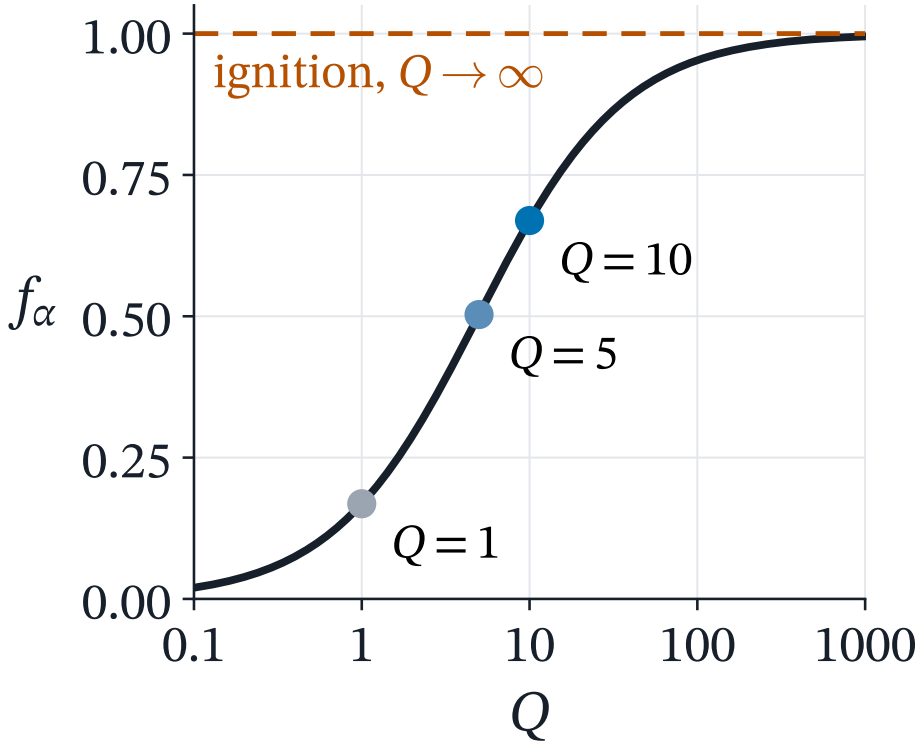


4.4 Power balance with gain

gain $Q = p_{\text{fus}}/p_{\text{ext}}$

$E_{\text{fus}} \approx 5E_\alpha$ $\quad f_\alpha = \dfrac{p_\alpha}{p_\alpha + p_{\text{ext}}} = \dfrac{Q}{Q+5}$

Q		f_α
1	break-even	1/6
5	burning plasma	1/2
∞	ignition	1



4.4 Triple product

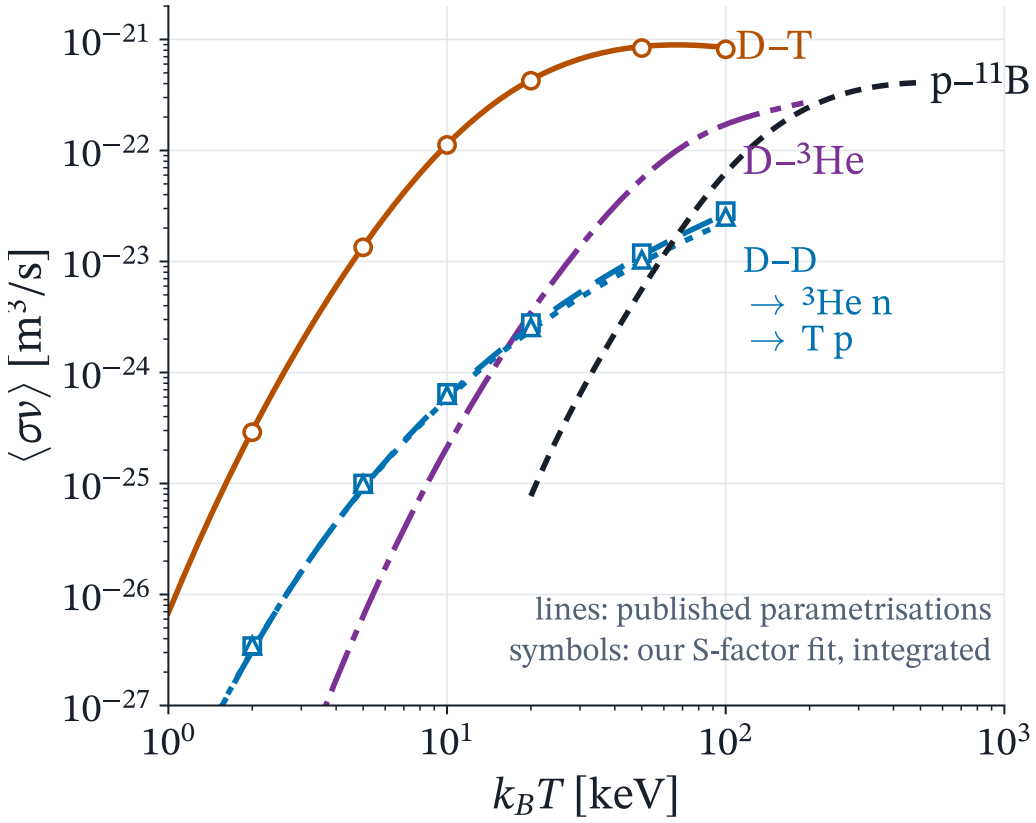
ignition $\frac{n^2}{4}\langle\sigma v\rangle E_\alpha = \frac{3nk_BT}{\tau_E}$

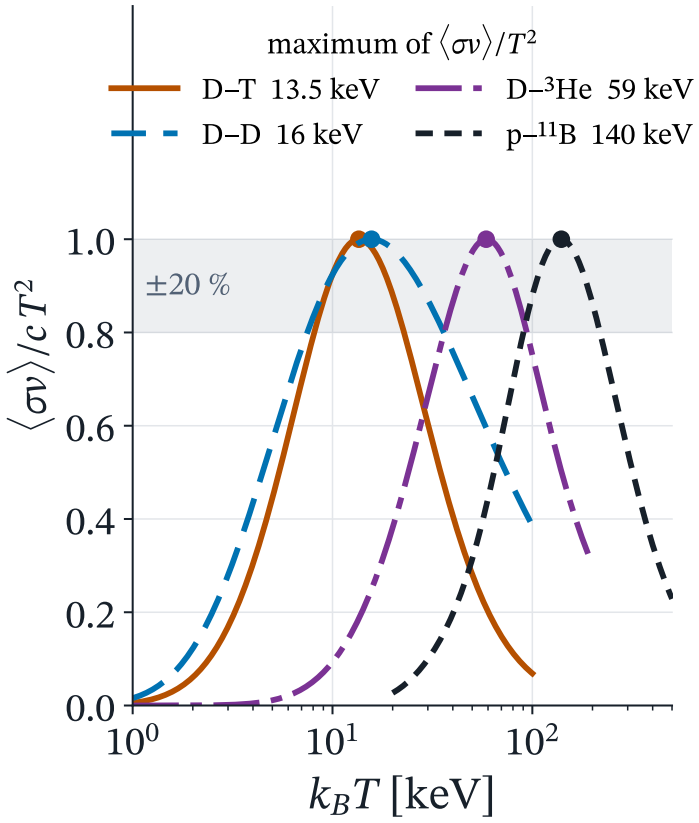
10–20 keV $\langle\sigma v\rangle \propto T^2$ so $nT\tau_E$ is constant

$$nk_BT\tau_E = \frac{12(k_BT)^2}{\langle\sigma v\rangle E_\alpha} \approx 3\times10^{21}\,\mathrm{keV\,s\,m^{-3}}$$

= 5 bar s ion pressure, 10 bar s ion + electron pressure

4.4 How good is $\langle \sigma v \rangle \approx c T^2$?



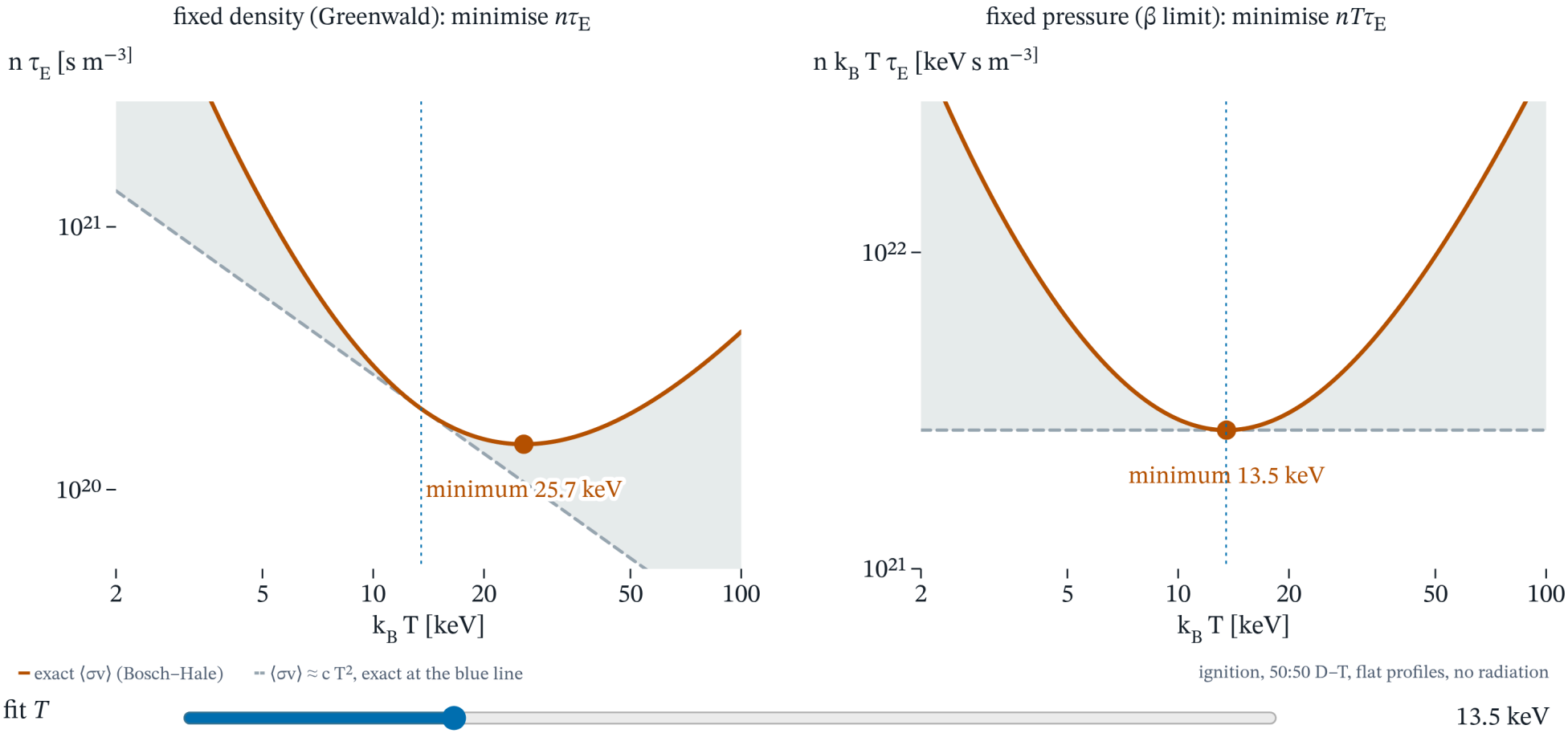


within 20 %: **D–T 8–23 keV** D–D 7.5–36 keV D–^{3}He 36–94 keV p–^{11}B 91–218 keV

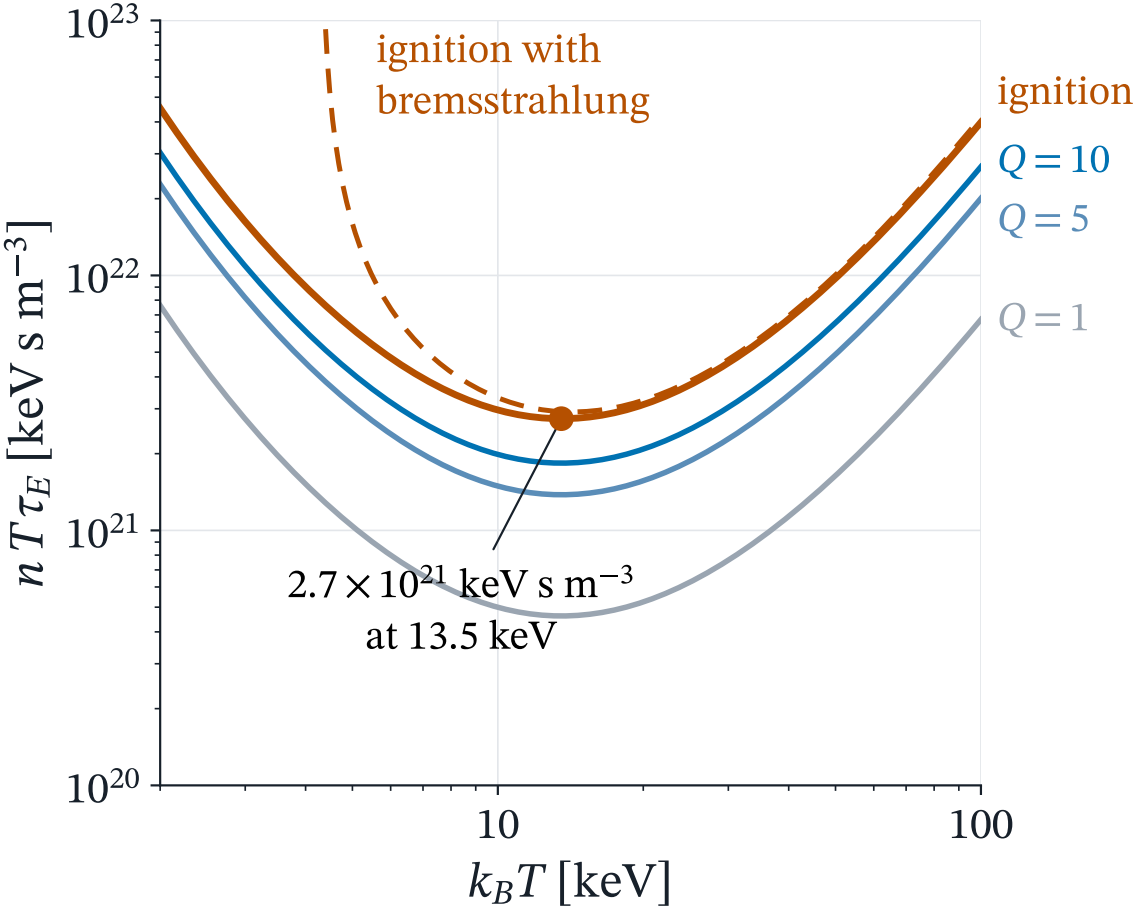
c: maximum of $\langle \sigma v \rangle / T^2$ for each fuel (grey band)

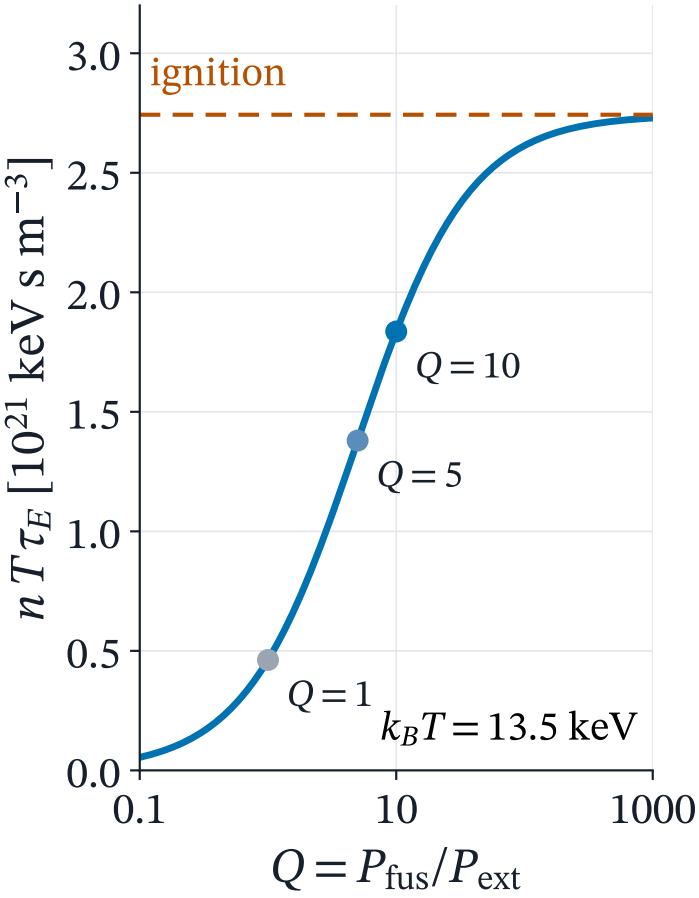
Bosch & Hale · Nucl. Fusion 32, 611 (1992) · Nevins & Swain · Nucl. Fusion 40, 865 (2000)

4.4 $n\tau_E$ or $nT\tau_E$?



4.4 Lawson plane



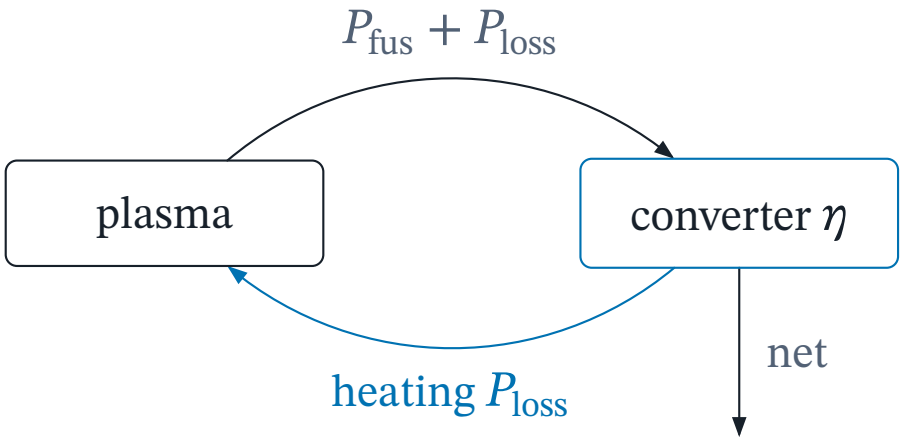


dashed: with bremsstrahlung (4.9)

Bosch & Hale · Nucl. Fusion 32, 611 (1992) · notebook: Lawson criterion

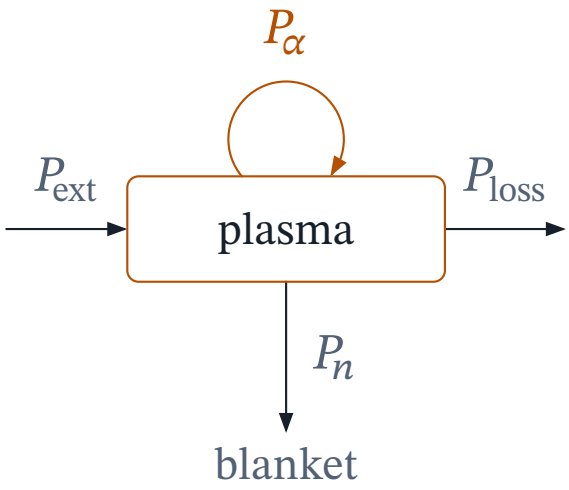
4.5 Lawson 1957 and self-heating

Lawson 1957



$$P_{\text{fus}} \geq 2P_{\text{loss}} \quad (\eta = 1/3)$$

Self-heating



$$P_\alpha \geq P_{\text{loss}} \quad \text{(ignition)}$$

Lawson · Proc. Phys. Soc. B 70, 6 (1957) · Wurzel & Hsu · Phys. Plasmas 29, 062103 (2022)

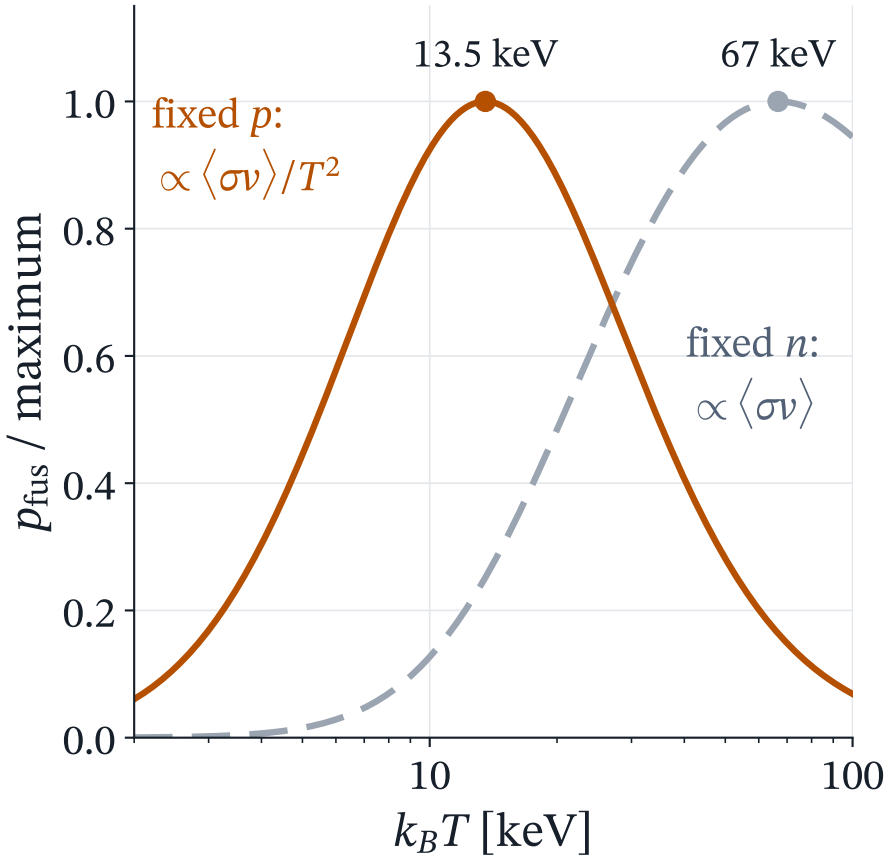
4.6 Fixed pressure

pressure $p = 2nk_BT, \quad \beta = \dfrac{2\mu_0 p}{B^2}$

$$p_{\text{fus}} \propto \frac{\langle \sigma v \rangle}{(k_B T)^2} \beta^2 B^4$$

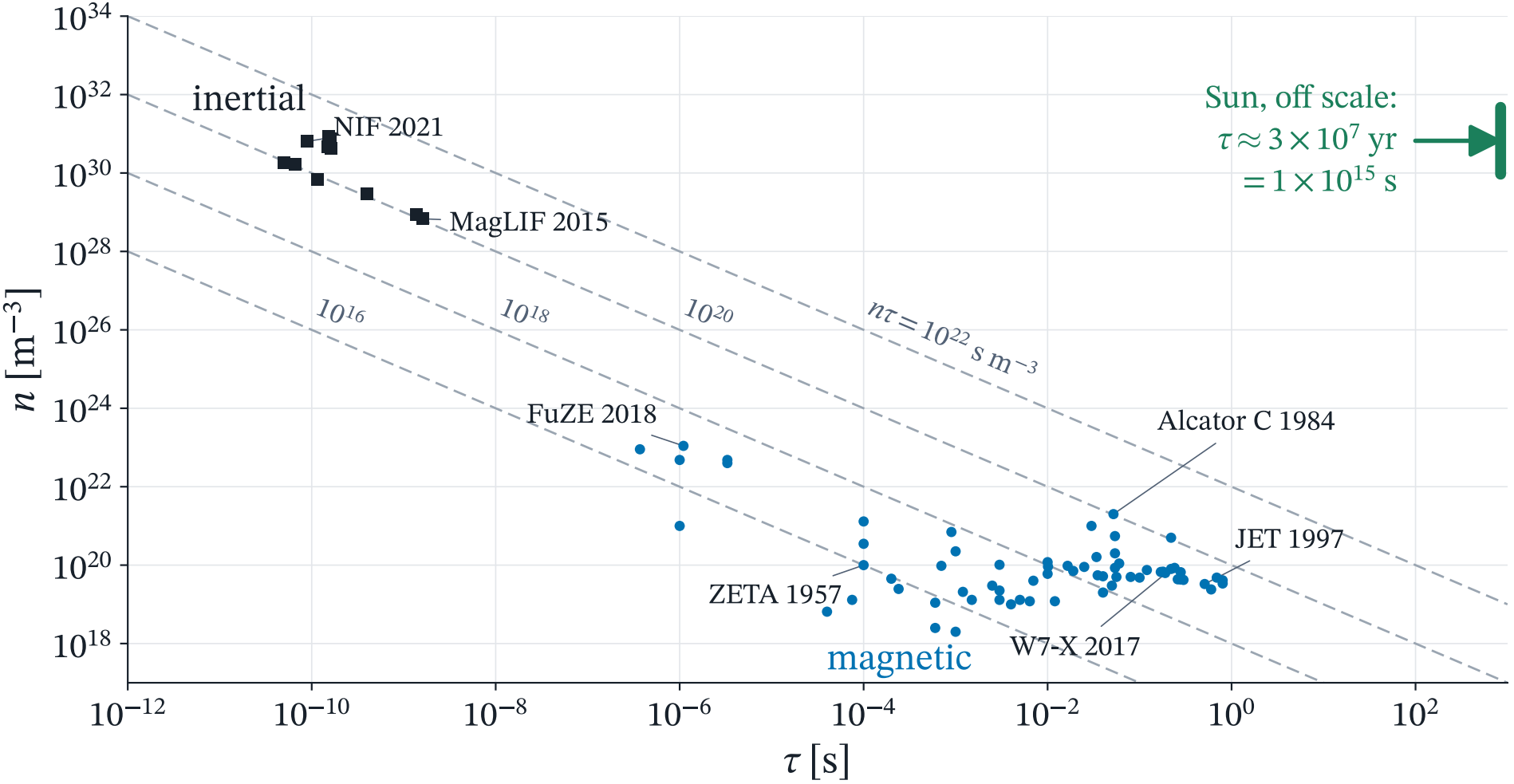
maximum of $\langle \sigma v \rangle / T^2$: 13.5 keV

density $n_G = I_p/(\pi a^2)$ $10^{20}\,\text{m}^{-3}$, MA, m



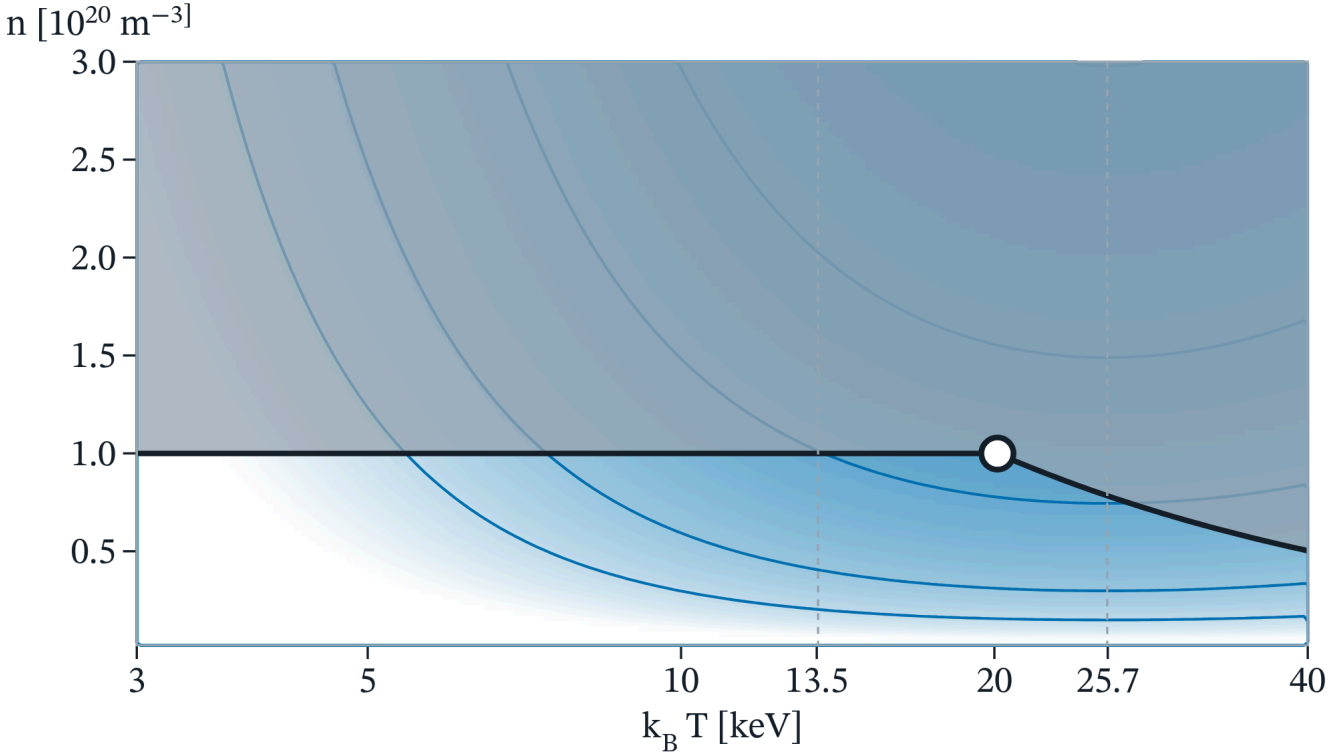
Greenwald · Plasma Phys. Control. Fusion 44, R27 (2002)

4.7 Density and confinement time



Data: Wurzel & Hsu · Phys. Plasmas 29, 062103 (2022), Tables 6–8, CC BY 4.0

4.8 Density and pressure limits



τ_E needed [s]

0.3 1 3 10

blue lines: $\tau_E = 0.5, 1, 2, 5, 10$ s

grey: beyond density or β limit

dashed: 13.5 keV (best at fixed p), 25.7 keV (best at fixed n)

least τ_E needed ○

$k_B T = 20.1$ keV, $n = 1.00 \times 10^{20}$ m^{-3}

$\tau_E = 1.55$ s, $p = 6.4$ bar

limits cross at $T_\times = p_{max}/2n_{max} = 20.1$ keV

optimum = $T_\times$ clamped to [13.5, 25.7] keV

B [T] — 6.0 T

β_{max} [%] — 4.5 %

n_{max} [10^{20}] — 1.00×10^{20} m^{-3}

target ○ Q = 1 ○ Q = 10 ◉ ignition

no density limit (stellarator) ☐

4.8 Limits and confinement scaling

limits $n \le n_G, \quad p \le \beta_{\max} B^2/2\mu_0 \quad \Rightarrow \quad T_\times = p_{\max}/2n_{\max}$

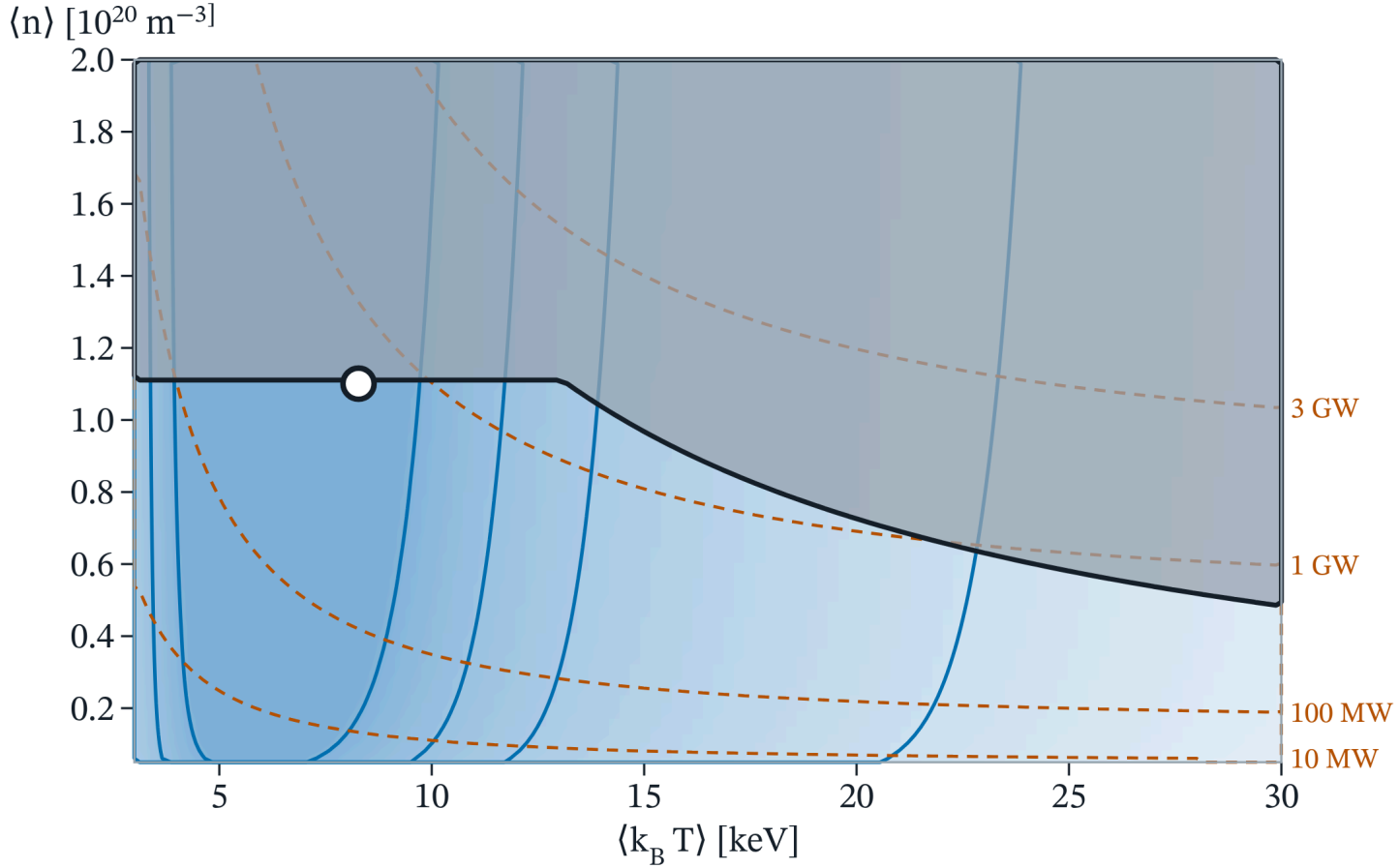
tokamak $T_\times \approx 0.39\,\beta_N a B/f_{\mathrm{GW}}$ keV ITER 8.8 keV

IPB98(y,2) $\tau_E \propto I^{0.93} R^{1.97} \overline{n}^{0.41} P^{-0.69} \ldots$

τ_E falls with heating power $\Rightarrow$ optimum $\approx$ 6–9 keV

ITER Physics Basis · Nucl. Fusion 39, 2175 (1999) · Yamada et al. · Nucl. Fusion 45, 1684 (2005)

4.8 Operating space of a device



gain Q

0.1 · 1 · 5 · 10 · 30

blue lines: $Q = 1, 5, 10, 30$ · orange: P_{fus} [MW]

grey: beyond Greenwald or β_N limit

ignited, highest P_{fus} ○

$k_B T = 8.3$ keV, $n = 1.10 \times 10^{20}$ m^{-3}

$Q = \infty$, $P_{fus} = 688$ MW

$\tau_E = 3.05$ s, $\beta = 2.7$ %

$P_{ext} = 0$ MW, $P_\alpha = 139$ MW

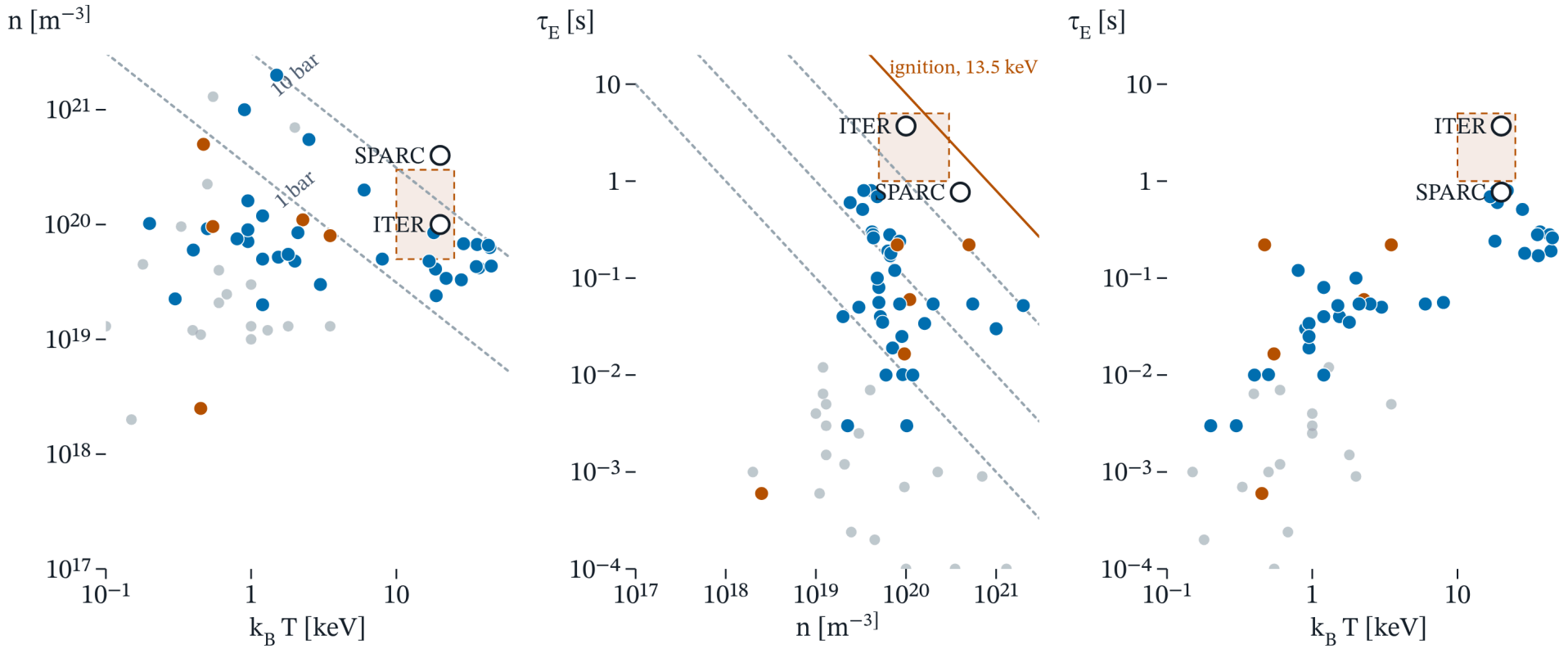
0-D, parabolic profiles ($\alpha_n = 0.2$, $\alpha_T = 1$), fuel 85 % of n_e, axes volume averages, $T_e = T_i$, bremsstrahlung only; τ_E = H·IPB98(y,2) or H·ISS04 with $P = W/\tau_E$ solved self-consistently. Trends, not design values.

device: ITER (tokamak)

B [T]: 5.3 T

H factor: 1.00

4.8 Where present machines stand



● tokamak ● stellarator ● other magnetic ○ design ▭ D–T reactor window

Peak values; gain curves for peaked profiles, Z_{eff} 1.5–3.4; data: Wurzel & Hsu, Phys. Plasmas 29, 062103 (2022)

view ◉ three cross-sections ○ triple product

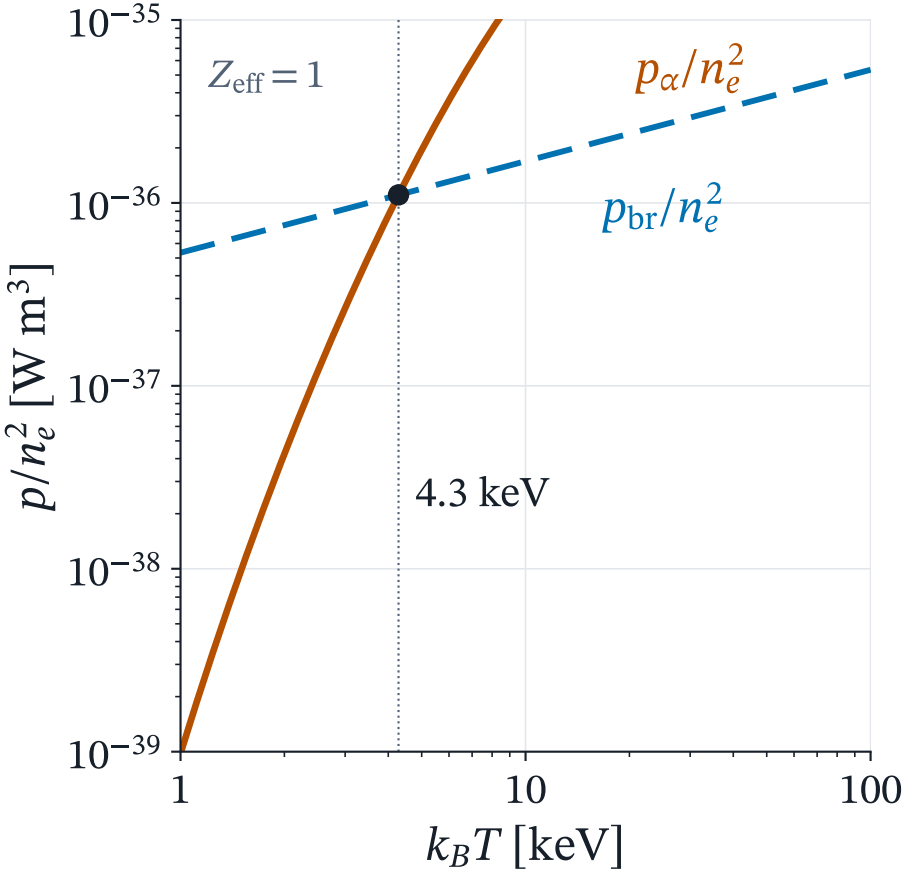
4.9 Bremsstrahlung and ideal ignition

electron–ion $p_{\mathrm{br}} = C_B n_e^2 Z_{\mathrm{eff}} \sqrt{k_B T}$

$p_\alpha = p_{\mathrm{br}}$ $\langle \sigma v \rangle E_\alpha = 4 C_B Z_{\mathrm{eff}} \sqrt{k_B T}$ n cancels

$$k_B T_{\mathrm{ideal}} = 4.3\ \mathrm{keV}$$

D–T, $Z_{\mathrm{eff}} = 1$, any density



$C_B = 5.34 \times 10^{-37}\ \mathrm{W\ m^3 keV^{-1/2}}$

Lawson · Proc. Phys. Soc. B 70, 6 (1957) · NRL Plasma Formulary

4.9 Cyclotron radiation

one electron $$P = \frac{e^4 B^2 v_\perp^2}{6\pi\varepsilon_0 m_e^2 c^3}$$

Maxwellian $$p_{\mathrm{cyc}} \approx 6.2 \times 10^{-17}\, n_e\, k_B T_e\, B^2 \ \mathrm{W\,m^{-3}}$$

$10^{20}\,\mathrm{m^{-3}}$, 10 keV, 5 T [MW m^{-3}]:

p_{fus}	p_{br}	p_{cyc} emitted
0.8	0.017	1.6

optically thick: mostly reabsorbed, walls reflect

grows with T_e and B: a main loss for D–^{3}He and p–^{11}B

n_e in m^{-3}, $k_B T_e$ in keV, B in T pairs need $2m_e c^2 = 1.02$ MeV: negligible for $k_B T \ll 511$ keV

NRL Plasma Formulary

4.10 Quasineutrality and dilution

quasineutral $n_e = n_\mathrm{D} + n_\mathrm{T} + \sum_Z Z n_Z$

concentration $c_Z = n_Z / n_e$

fixed n_e $p_\mathrm{fus} \propto \left(1 - \sum_Z Z c_Z\right)^2$

radiation $Z_\mathrm{eff} = 1 + \sum_Z Z(Z-1) c_Z$

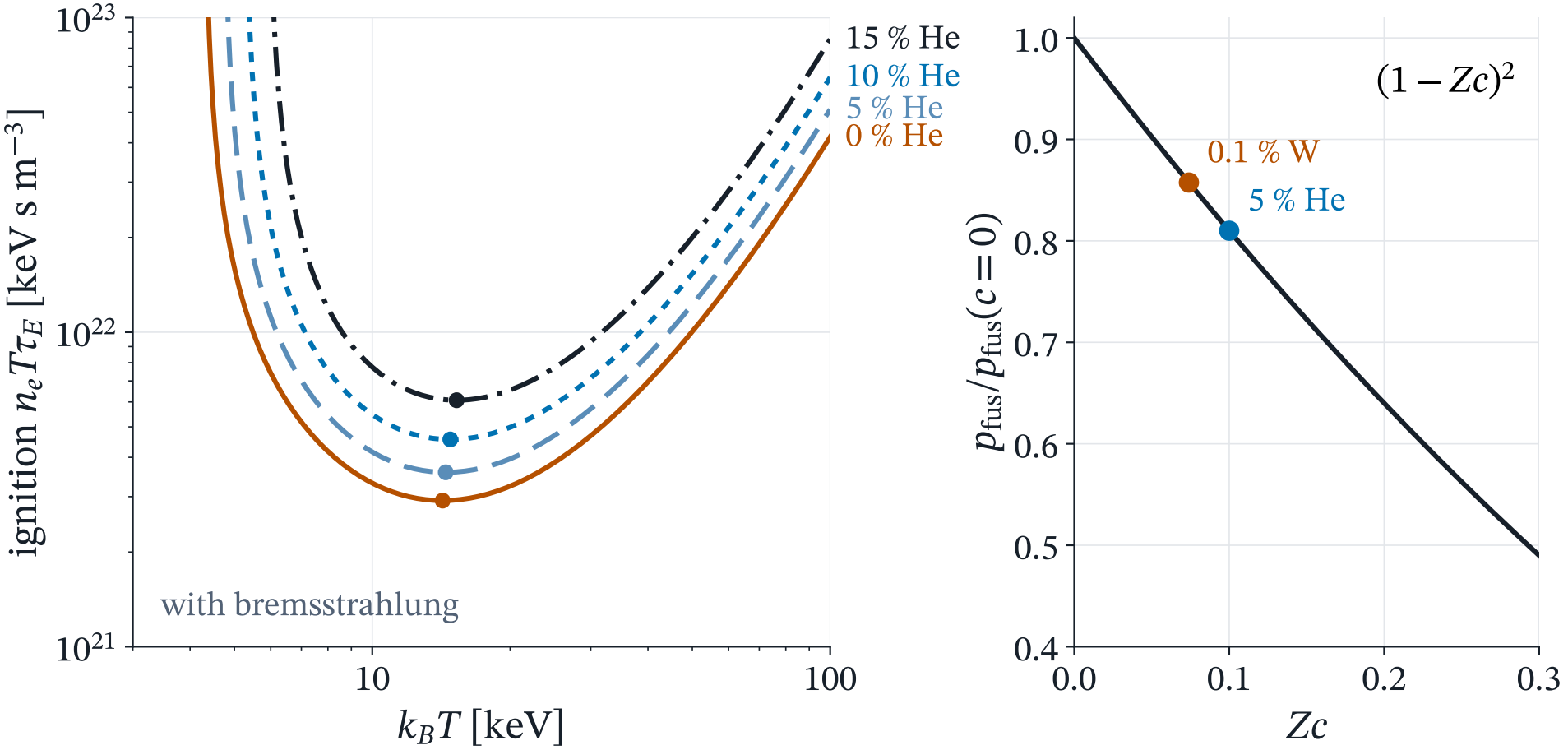
	Zc	p_fus
0.1 % W	0.074	0.86
5 % He	0.10	0.81

one W ion displaces 74 fuel ions

W radiates lines: $c_W = 3 \times 10^{-5}$ raises the minimum $nT\tau_E$ by 20 %

Pütterich et al. · Nucl. Fusion 50, 025012 (2010)

4.10 Helium ash



helium concentration c: fuel $1-2c$, $Z_{\mathrm{eff}} = 1+2c$

notebook: impurities and fuel dilution